DATA SCIENCE | DS-A4 | HOMEWORK

Predicting adult Red Knot sex from morphometric data

Model Comparison and Calibration Lab

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Research question

How do logistic regression, random forest and gradient boosting compare in discriminating between male and female adult Red Knots and producing reliable probabilities, and how should a decision threshold be selected under explicit error costs and resource constraints?

1. Data and analytical sample

Adult Red Knots (*Calidris canutus canutus*) captured at Banc d'Arguin, Mauritania, from the dataset accompanying Oortwijn et al. (2025) [1]. Reference sex was determined by DNA analysis. Predictors are wing, bill and diagonal tarsus lengths, all in millimetres. Female is the positive class (F=1; M=0).

Preparation step	Records / individuals
Original file	2,165 records; 2,131 ring identifiers
Missing values / identical rows	0 / 0
Conflicting reference sex	Both records for Z062234 excluded
Repeated captures	33 later records excluded after conflict removal
Analytical sample	2,130 unique birds: 1,263 F and 867 M

The earliest capture per ring was retained, ordered by Year and Month; original row order breaks within-month ties. Exclusions do not depend on morphometric extremeness. Ring, Year and Month are not predictors.

Evaluation design

Stage	Training birds	Validation birds	Purpose
Outer CV: 5 folds	1,704	426	Evaluate each tuned procedure
Inner CV: 3 folds	1,136	568	Select model settings

All model families use identical stratified outer and inner folds, with shuffle=True and random_state=42. Each bird receives one outer-fold probability per model. Learned preprocessing is fitted within the relevant training subset. Model settings and thresholds use only outer-training data.

Data reuse: the dataset and earlier test results were already examined in DS-A3. The present procedure separates fitting and evaluation within DS-A4; it is not validation on a new independent dataset.

2. Models and predictive performance

Model	Fixed settings	Six tuning configurations
Logistic regression (LR)	StandardScaler pipeline; L2; lbfgs; max_iter=5,000	C = 0.01, 0.1, 1, 10, 100, 1000
Random forest (RF)	300 trees; bootstrap; max_features=sqrt	max_depth = 3, 6, None; min_samples_leaf = 5, 15
Gradient boosting (GB)	learning_rate=0.05; min_samples_leaf=5; log loss	n_estimators = 50, 100, 200; max_depth = 1, 2

All candidates use Wing, Bill and Tarsus, without class weighting or resampling. Each family receives six configurations on the same three inner folds: 18 search fits per outer fold. The budget is equal configuration counts, not equal runtime. Mean inner-validation Brier score selects the settings.

Probability quality and discrimination

Model	Brier score	ECE	ROC-AUC	Average precision
Logistic regression	0.1586 [0.1501, 0.1669]	0.0229 [0.0184, 0.0436]	0.8412 [0.8251, 0.8580]	0.8852 [0.8718, 0.8986]
Random forest	0.1620 [0.1538, 0.1696]	0.0230 [0.0191, 0.0451]	0.8349 [0.8184, 0.8524]	0.8784 [0.8639, 0.8932]
Gradient boosting	0.1604 [0.1520, 0.1685]	0.0280 [0.0211, 0.0477]	0.8370 [0.8206, 0.8540]	0.8769 [0.8611, 0.8923]

Values are pooled outer-fold estimates, with approximate 95% bootstrap intervals in brackets. Lower Brier and ECE are better; higher ROC-AUC and AP are better. ECE uses 10 equal-width bins. Probabilities are assessed without an additional post-hoc calibration step.

Reference classification at P(female) >= 0.50

Model	Accuracy (%)	Balanced accuracy (%)	Precision F (%)	Recall F (%)	Recall M (%)
LR	76.67	75.12	78.54	83.45	66.78
RF	75.73	73.89	77.23	83.77	64.01
GB	76.81	75.29	78.72	83.45	67.13

LR correctly classifies 1,633 birds; GB classifies 1,636 and RF 1,613. GB improves accuracy over LR by only three birds. LR has the lowest observed Brier score and the highest ROC-AUC and AP.

Paired Brier differences

Difference: A - B	Estimate	Approximate 95% interval
LR - RF	-0.003319	[-0.005859, -0.000683]
LR - GB	-0.001753	[-0.003512, 0.000062]
RF - GB	0.001566	[-0.000337, 0.003372]

Negative differences favour A. The LR-RF interval is entirely below zero; the LR-GB interval includes zero. Intervals including zero do not establish equivalence.

Uncertainty uses 2,000 sex-stratified paired percentile bootstrap resamples, seed=42. Each iteration samples the same birds for every model. Models are not refitted: intervals are conditional descriptive summaries and do not fully capture training, split or overlapping-CV dependence.

3. Discrimination and calibration

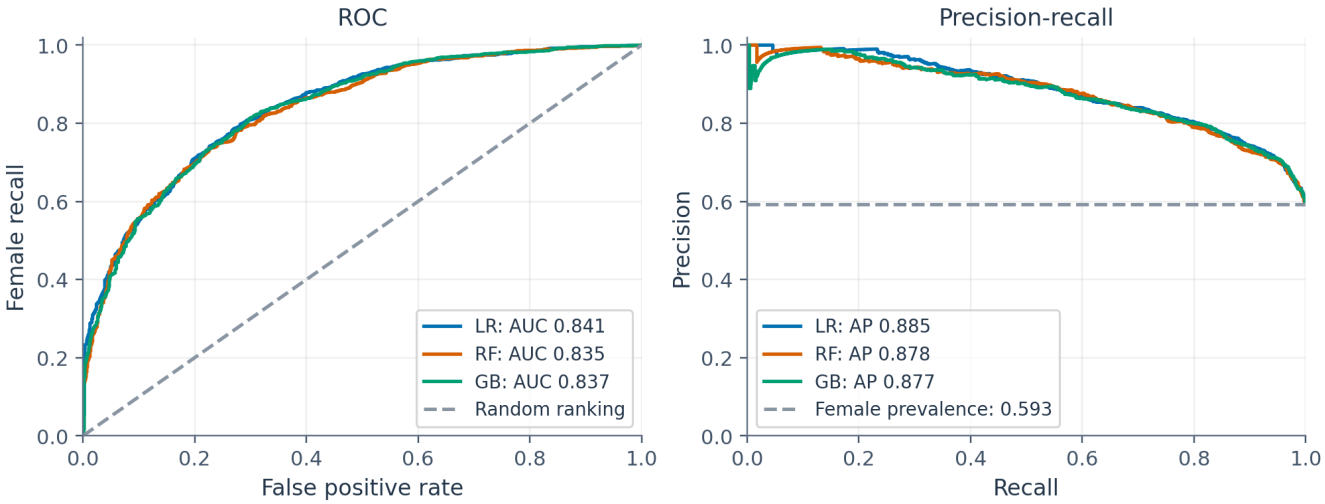


Figure 1. ROC and precision-recall curves from paired outer-fold predictions. The PR reference is the observed female prevalence (0.593). LR, RF and GB abbreviations are defined on page 2.

The curves largely overlap. All three models discriminate between the classes, while the additional complexity of RF and GB produces no consistent observed improvement. High precision at low recall corresponds to selecting a smaller subset of likely females; precision decreases when more females are sought.

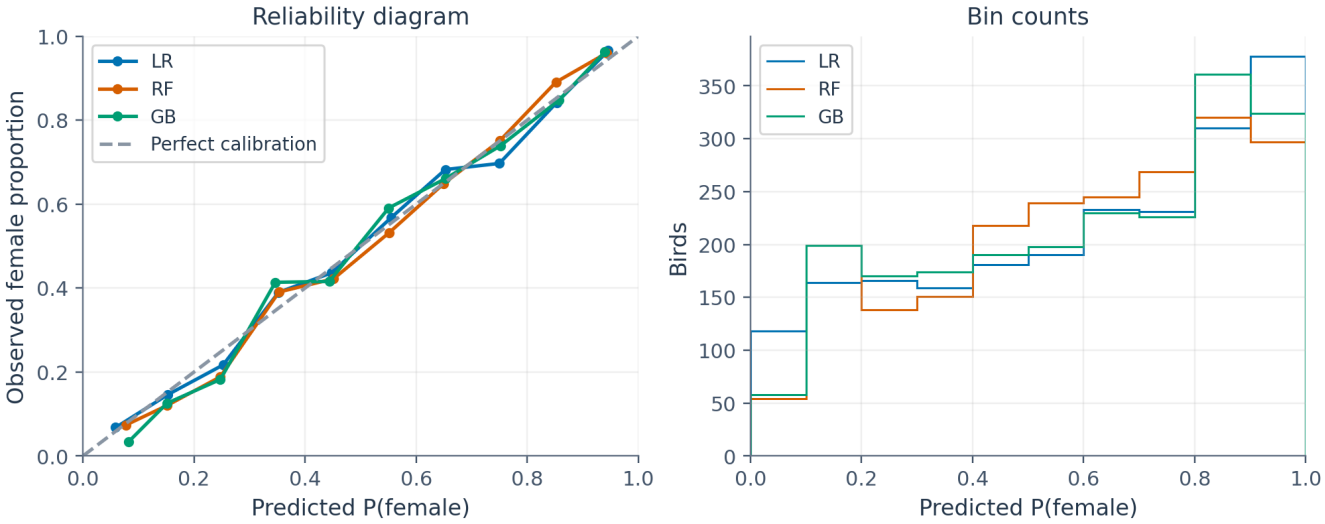


Figure 2. Reliability diagram and prediction counts using the same 10 equal-width bins as ECE. Curves are displayed only for nonempty bins; the diagonal denotes perfect calibration.

Observed female proportions broadly follow predicted probabilities. Local deviations remain: for LR, the bin near predicted probability 0.75 has an observed female proportion near 0.70. Bin-specific uncertainty intervals are not shown, so such deviations should not be treated as definitive systematic miscalibration.

Interpretation. Good grouped calibration is compatible with individual mistakes. A well-calibrated prediction of 0.70 still allows about 30% males among similarly scored birds. ROC-AUC, calibration and threshold-based accuracy describe different aspects of performance.

4. Error slices and failure cases

The following exploratory diagnostics were specified after inspecting aggregate performance. They do not change model settings, decision rules or the analytical sample. Classification uses the reference threshold of 0.50.

Reference-sex groups

Model	Reference sex	Birds	Brier	Error rate (%)	Recall (%)
LR	F	1,263	0.1200	16.55	83.45
LR	M	867	0.2150	33.22	66.78
RF	F	1,263	0.1210	16.23	83.77
RF	M	867	0.2216	35.99	64.01
GB	F	1,263	0.1209	16.55	83.45
GB	M	867	0.2179	32.87	67.13

Errors are more frequent among males. LR misclassifies 33.22% of males versus 16.55% of females; the same asymmetry is present in RF and GB. Calibration is not assessed within single-sex groups.

Bill-length groups

Model	Bill range	Birds	Brier	Error rate (%)	Male recall (%)
LR	Central	1,102	0.1906	29.04	52.26
LR	Outer	1,028	0.1244	17.22	80.49
RF	Central	1,102	0.1959	29.67	51.31
RF	Outer	1,028	0.1256	18.48	76.01
GB	Central	1,102	0.1926	28.49	54.87
GB	Outer	1,028	0.1258	17.51	78.70

Central means the inclusive 25th-75th percentile range of Bill in the corresponding outer-training subset, applied identically to all models. LR error is 29.04% centrally versus 17.22% outside; male recall drops to 52.26%. These associations do not establish a biological cause.

High-confidence errors

Model	Confidence >= 0.90: birds	Errors	Error rate (%)
LR	496	21	4.23
RF	351	16	4.56
GB	382	14	3.66

Ring	DNA sex	Predicted	P(female)	Wing	Bill	Tarsus
Z062396	F	M	0.0098	160	30.5	30.2
Z049413	M	F	0.9782	178	38.9	31.3
Z063154	M	F	0.9755	178	38.7	34.9

Confidence is max[p, 1-p]. High-confidence subsets differ between models; their error rates are not paired subgroup comparisons. Confident errors neither prove reference-label errors nor by themselves demonstrate poor overall calibration. These birds remain in the analysis.

Counterexample to universal superiority: LR has the best overall Brier, yet GB has a lower observed central-range classification error (28.49% versus 29.04%). The small subgroup difference is descriptive, not established superiority.

5. Thresholds, cost and human capacity

The hypothetical task is to select likely females for DNA confirmation. Selecting a male costs 1 unit (FP), and not selecting a female costs 2 units (FN). Correct decisions have zero loss. Total loss is FP + 2 x FN. These are illustrative assumptions, not measured project costs.

At most 40% of each batch can be checked, rounding down. Birds must meet the threshold; if eligibility exceeds capacity, the highest probabilities are retained, with Ring in ascending order breaking ties. Non-selection does not assert male sex.

Thresholds selected within outer-training data

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
LR	0.73	0.71	0.72	0.72	0.73
RF	0.72	0.71	0.71	0.70	0.70
GB	0.73	0.72	0.73	0.72	0.72

Internal out-of-fold predictions for the selected configuration choose thresholds from 0.00-1.00 (step 0.01), plus select-none. Minimum loss wins; ties prefer fewer selections, then the higher threshold. These internal predictions are tuning data, not independent performance estimates.

Reference decision results: outer validation

Model	Selected	True F	FP	FN	Precision (%)	Female recall (%)	Loss per 100 birds
LR	823	729	94	534	88.58	57.72	54.5540
RF	825	732	93	531	88.73	57.96	54.2254
GB	820	723	97	540	88.17	57.24	55.2582

Selection fractions are 38.64% (LR), 38.73% (RF) and 38.50% (GB). Capacity can remain unused when fewer birds meet the threshold. Limits are enforced per outer-validation batch, without using its reference labels for selection.

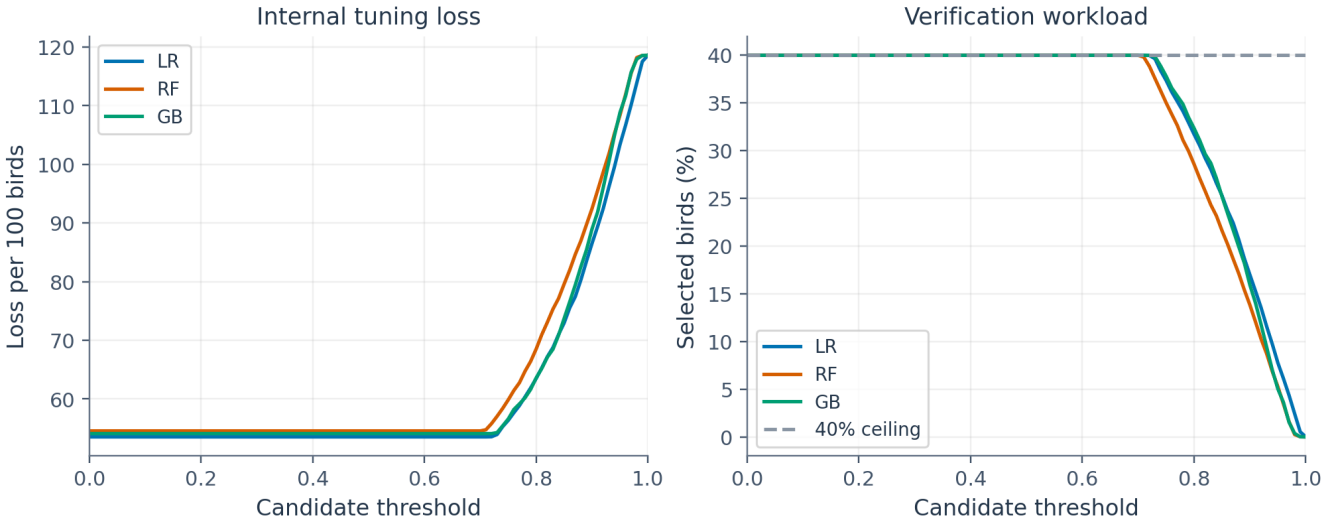


Figure 3. Reference-scenario tuning loss and workload, averaged over the five outer-training subsets. These are internal selection curves, not external performance curves.

The lower-threshold plateau reflects the capacity rule selecting the same highest-ranked birds. Beyond approximately 0.70, fewer birds are selected and missed females increase loss. RF has the lowest observed reference loss: 54.2254 versus 54.5540 for LR, a small difference of 0.3286 units per 100 birds.

6. Sensitivity, final decision and limits

FP cost remains 1; FN cost is varied across 0.5, 1, 2 and 5, and capacity across 20%, 40% and 60%. Each rule is selected internally and assessed on the same outer folds. Compare model losses within a scenario; changing cost assumptions changes the valuation scale.

FN cost	Capacity	LR loss	RF loss	GB loss	Lowest loss
0.5	20%	20.939	21.502	21.526	LR
0.5	40%	17.066	16.901	17.277	RF
0.5	60%	17.371	16.925	17.089	RF
1	20%	41.080	41.972	41.925	LR
1	40%	29.718	29.296	29.906	RF
1	60%	23.521	24.366	23.803	LR
2	20%	81.362	82.911	82.723	LR
2	40%	54.554	54.225	55.258	RF
2	60%	35.446	36.573	35.681	LR
5	20%	202.207	205.728	205.117	LR
5	40%	129.765	129.014	131.315	RF
5	60%	71.080	73.615	71.315	LR

Loss is per 100 birds. LR is lowest in all 20%-capacity scenarios; RF is lowest in all 40%-capacity scenarios. At 60%, RF is lowest for FN cost 0.5 and LR for the other costs. These related scenarios are descriptive, not independent replications.

Final decision record. Retain LR as the primary probability model by the declared Brier criterion. Final all-data tuning selects **C=0.1** and **P(female) >= 0.71**. Apply the 40% ceiling and ranking/tie rules. This finalization fit is not a new performance test; the exact fitted model and cutoff require external validation.

Claim boundary and limitations

Morphometrics provide useful but incomplete sex information in these adult birds. Additional complexity does not consistently improve probability quality or decision utility. Male identification and central bill lengths remain difficult, and confident individual errors occur.

The dataset was previously explored in DS-A3. Random folds do not demonstrate transfer to juveniles, new populations, sites or future periods. Bootstrap intervals are conditional, without refitting; ECE depends on binning and local calibration uncertainty is not shown. Subgroup analyses are exploratory. Costs and capacity are hypothetical, and lower observed scenario loss is not established deployment superiority.

Computational source and verification

Source notebook: `DS_A4_red_knot_calibration.ipynb`. The author ran the analysis in Colab (scikit-learn 1.6.1). A separate AI-assisted review reran all 19 code cells with the same scikit-learn version and adapted file paths; 13 saved numerical tables matched at their displayed precision, and C=0.1 / threshold=0.71 were reproduced. This was not an independent human review.

Input: `14._adults_Mauritania.csv` (69,868 bytes). SHA-256:

`6da0a6b44266f448393c92a293f24f145d803ee95b6246804a0185bbfd9869f6`

[1] Oortwijn, T., Lameris, T., Zhemchuzhnikov, M., et al. (2025). *Demand-resource mismatch explains body shrinkage in a migratory shorebird* [Dataset]. Dryad. Published 4 April 2025. https://doi.org/10.5061/dryad.3n5tb2rqb

Repository: github.com/kseniia-mamushkina/DS-A4-red-knot-calibration